Lecture 6 - Bayesian Learning

Each instance can increase or decrease the probability of a hypothesis being true. Bayesian Learning assigns each hypothesis a probability of being true based on its training instances (e.g. this patient has a 93% chance of getting cancer). This does require initial knowledge of many probabilities and has a high computational cost.

Given: instance space X and **discrete** class variable Y, with some unknown joint probability distribution $p(x, y)$, and training data D that is **iid** sampled from $X \times Y$. sampling is **independent** so we don't have to model dependencies between instances, and **identical** so the distribution of the data is the same.

Find: the most probable classification $y \in Y$ for a new instance according to $p(x, y)$.

Maximum A Posteriori Hypothesis (MAP)

Given a set of hypothesis' H, find the most probable hypothesis $h \in H$ given observed data D. This hypothesis is called **maximum a posteriori hypothesis**.

$$h_{MAP} = argmax\ p(D|h)p(h)$$

If $p(h)$ is uniformly distributed, then

$$h_{MAP} = argmax\ p(D|h)$$

Defining Probabilities

$p(h) = \frac{1}{|H|}$ (uniform distribution)

$p(D|h) = 1$ if h is **consistent** with D (i.e. h correctly classifies all the instances in D), otherwise 0

$p(D) = \sum_{h \in H} p(D|h)p(h) = \frac{|VS|}{|H|}$

Thus, if h is consistent with D then:

$$p(h|D) = \frac{1}{|VS|}$$

This means version spaces only contain MAP hypothesis'. Any consistent learner outputs an MAP hypothesis.

"Find the most probable class of this instance":

$$y = argmax \sum_{h \in H} p(y|h)p(h|D)$$

No other classification method using the same hypothesis space and same prior knowledge can outperform this classifier on average, but not the Bayes optimal classifier may not be in H.

Gibbs Classifier

1) Choose hypothesis at random according to $p(h|D)$

2) Use this hypothesis to classify new instance

Surprising fact: $E(error_{gibbs}) \leq 2E(error_{bayesoptimal})$

Bayes Rule

To solve the classification problem for a new instance, we need to know:

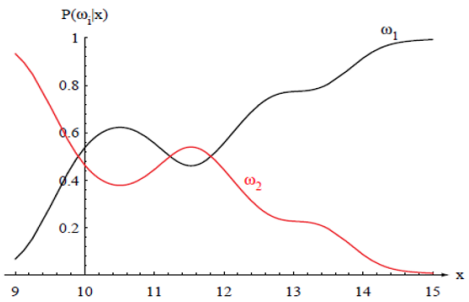
- the prior distribution $p(y)$

- the likelihood $p(x|y)$ for every $y \in Y$

- the evidence distribution $p(x)$

We can calculate the **posterior class probability** using:

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}$$



The winning class is the $argmax$ of $p(y|x)$

Version Spaces

Given a **finite** hypothesis space H and training data D, version space $VS(D)$ is the set of hypothesis' that are observed at least once in the training data. Formally:

$$VS(D) = h \in H : (\forall (x, y) \in D : y = h(x))$$

Minimum Description Length (MDL)

$$h_{MDL} = h_{MAP} = argmin\ L_{c1}(h) + L_{c2}(D|h)$$

Where $L_{c1}(h)$ is the description length of h:

$$-log_2(p(h))$$

And $L_{c2}(D|h)$ is the description length of D under hypothesis h:

$$-log_2(p(D|h))$$

This means we have to choose the hypothesis the minimizes the sum of the description lengths.

Naive Bayes Classifier

The naive Bayes classifier naively assumes that the input variables are conditionally independent. This classifier does not emplot any search in the hypothesis space, it just multiplies probability estimates:

$$\hat{y} = argmax \prod_{i=1}^{p} p(x_i|y)p(y)$$

Estimating $p(x_i|y)$ for discrete variables:

- Use relative frequency, m-estimated accuracy etc.

Estimating $p(x_i|y)$ for continuous variables: Gaussian Normal Density:

$$\frac{1}{\sigma_i \sqrt{2\pi}} e^{-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}}$$